AUTONOMOUS LINE-FOLLOWING WAITER ROBOT

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BE MECHATRONICS (2024-2028)

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Contents

Team Member Word Count & Contribution Matrix

Team Member Name	Role	Word Count (in Assigned Color Code)
Adeel Murad	Team Lead	2000
Hamaad Naeem	Project Manager & Tech	3000
Muneeb Ur Rehman	Integrator and Tester	2500

Chapter 1- Preliminaries:

1.0 Proposal

The proposed project involves the design and development of an Intelligent Autonomous Waiter Robot based on an ATMEL microcontroller platform. The system can perform autonomous multi-point delivery and return operations within a predefined indoor environment. The robot follows black-line pathways, navigates toward assigned customer tables, identifies the destination using computer vision techniques, and autonomously returns to its starting position after delivery completion.

The system integrates an ATMEL controller as the primary processing unit, an ESP32-CAM module for Edge AI-based target recognition, wheel encoders for distance measurement, an IMU sensor for orientation tracking, an SD card module for route storage, wireless communication modules for telemetry transmission, and motor driver circuitry for motion control.

The robot operates by receiving a delivery destination, loading the corresponding route from memory, and navigating through a line network while continuously monitoring its position. Upon reaching the designated destination, the ESP32-CAM verifies the target table using visual identification techniques. After successful confirmation, the robot performs a controlled 180-degree rotation and executes a path-reversal algorithm to safely return to the Kitchen Station.

Real-time telemetry data including current location, battery voltage, battery current, delivery status, and system health are transmitted wirelessly to a Central Monitoring Station. The stored route information on the SD card enables reliable navigation and accurate path recall during return operations.

The proposed design demonstrates the practical implementation of embedded systems, finite state machines, computer vision, sensor fusion, wireless communication, and autonomous navigation technologies in a single integrated robotic platform.

1.1 Initial Feasibility

An initial feasibility study indicates strong potential for successful implementation of the proposed system. The required hardware components including ATMEL microcontrollers, ESP32-CAM modules, wheel encoders, IMU sensors, line sensors, SD card modules, motor drivers, and battery monitoring circuits are commercially available and supported by extensive technical documentation.

The project architecture follows a modular design approach, simplifying hardware integration, firmware development, testing, and troubleshooting. The navigation algorithm is based on a deterministic state machine structure, allowing predictable system behaviour and easier debugging. Potential challenges include reliable junction detection, accurate path memory management, encoder calibration, IMU drift compensation, visual identification accuracy, and synchronization between route storage and retrieval processes. These challenges can be effectively addressed through systematic testing, sensor calibration, filtering algorithms, and iterative firmware optimization.

Considering component availability, implementation complexity, and development resources, the project is technically feasible, economically viable, and suitable for practical realization within an academic embedded systems environment.

1.2 Specifications of Deliverables

The proposed Waiter Robot project will include the following deliverables:

- Complete ATMEL microcontroller-based embedded system.
- Proteus simulation of the entire robot control system.
- Custom PCB design developed using any suitable PCB design software.
- Autonomous line-following navigation system.
- Multi-point delivery capability to Tables B, C, D, and E.
- ESP32-CAM based Edge AI visual identification system.
- Path memory and route storage using Micro SD card.
- Path reversal and autonomous return-to-home functionality.
- Wheel encoder integration for distance and speed measurement.
- IMU integration for heading estimation and accurate 180° turning.
- Wireless telemetry communication with a Central Monitoring Station.
- Battery voltage and current monitoring circuitry.
- Start/Stop interrupt-based control mechanism.
- Real-time display of battery and operational status.
- Complete firmware utilizing GPIO, ADC, Timers, Interrupts, SPI, UART, PWM, and State Machine programming techniques.
- Operational battery-powered robot capable of continuous autonomous operation.

1.3 Comparison Table of Similar Products

The following comparison parameters highlight the market positioning and technical enhancements of the proposed prototype compared to standard existing options:

Feature / Product	Proposed Robot	Existing Product A	Existing Product B
Navigation Mechanism	Color/Line Tracking + IMU	Magnetic Strip Only	LiDAR Mapping
Cost Performance	Low (~20k PKR)	Medium (~80k PKR)	High (~350k PKR)
Primary Controller	ATmega2560 MCU	PIC Microcontroller	Single Board Computer (SBC)

1.4 Technical Standards

Operational Path Width:

Optimized for standard indoor structural lines of exactly 30 mm width layout.

Turning Mechanics:

Built for precise 180-degree zero-turn radius capability using synchronized Inertial Measurement Unit (IMU) digital feedback structures.

1.5 Team Roles & Details

Adeel Murad (Team Lead):

Primarily structural firmware development, closed-loop algorithmic synchronization, and hardware module integration.

Hamaad Naeem (Project Manager & Technical Support):

Focuses on mechanical chassis structural planning, CAD model design and sensor integration

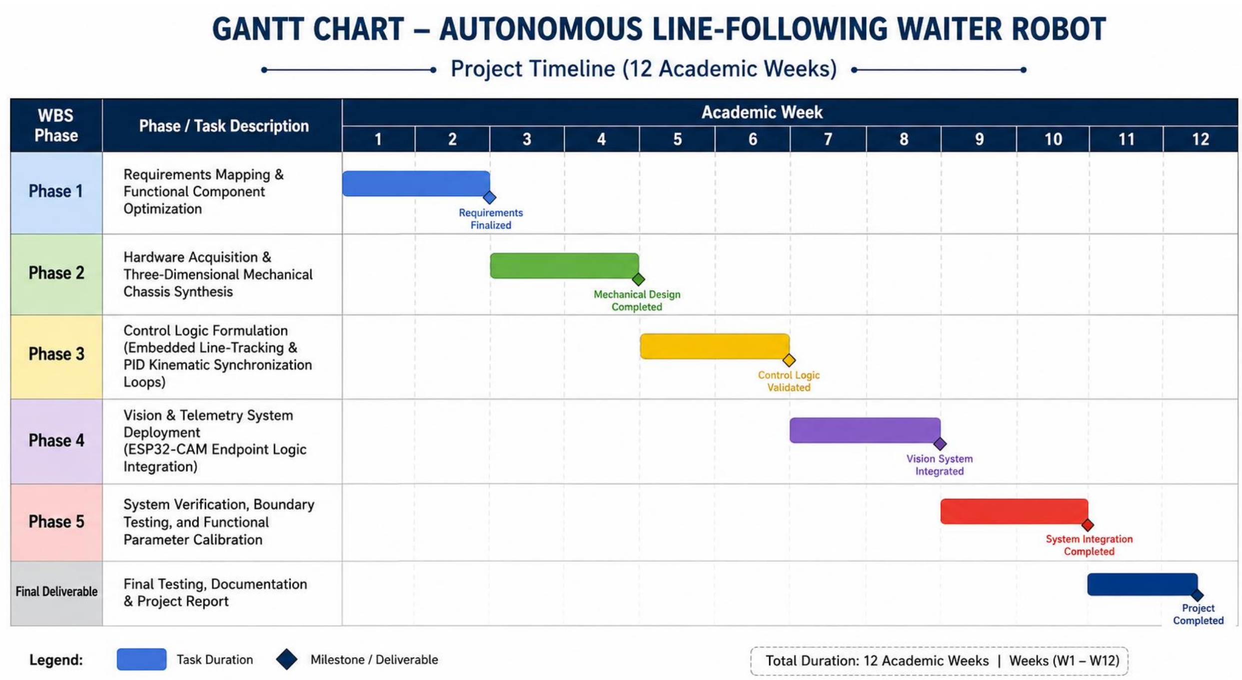
Muneeb Ur Rehman (Integrator & Tester):

Manages systemic wiring schematics, calibration parameters across reflective tracking sensor fields, debuging.

1.6 Work Breakdown Structure (WBS)

Phase 1: Requirements Mapping & Functional Component Optimization
Phase 2: Hardware Acquisition & Three-Dimensional Mechanical Chassis Synthesis
Phase 3: Control Logic Formulation (Embedded Line-Tracking & PID Kinematic Synchronization Loops)
Phase 4: Vision & Telemetry System Deployment (ESP32-CAM Endpoint Logic Integration)
Phase 5: System Verification, Boundary Testing, and Functional Parameter Calibration

1.7 Gantt Chart



The design timeline spans 12 academic weeks covering: requirements tracking (W1-2), mechanical CAD sourcing (W3-4), electronic circuit layout assembly (W5-6), embedded core script synthesis (W7-8), loop integration (W9-10), and laboratory verification checks (W11-12).

1.8 Estimated Budget

• Total Fiscal Allocation: 20,000 PKR
• Resource Distribution Matrix: Main control processing modules (35%), Actuation & Power Drivers (25%), Acrylic Structural Base Sourcing (20%), Batteries & Tracking Sensors (20%).

Chapter 2-Project Conception

Objective

The objective of this project is to design and develop an autonomous line-following waiter robot capable of performing intelligent multi-point delivery and return operations within a structured indoor environment. The robot shall autonomously navigate from a Kitchen Station to designated customer tables through predefined line paths, identify the correct destination using computer vision techniques, execute delivery confirmation procedures, and safely return to the original starting point using a path-memory and reversal algorithm.

The system will also provide wireless telemetry, battery monitoring, route storage, encoder feedback, IMU-based orientation tracking, and interrupt-driven user control, demonstrating the practical application of embedded systems and autonomous robotics.

2.1 Abstract

The Intelligent Waiter Robot is an autonomous embedded robotic platform designed to perform smart hospitality delivery operations. The robot follows predefined line networks to navigate between a Kitchen Station and multiple customer tables. Through the integration of an ATMEL microcontroller, ESP32-CAM vision module, wheel encoders, IMU sensors, SD card storage, and wireless communication technologies, the system achieves reliable navigation, destination identification, path memory, and autonomous return functionality.

The robot continuously monitors sensor data to make navigation decisions and maintain accurate positioning. Route information is stored on an SD card and recalled when required for return navigation. The ESP32-CAM module performs Edge AI-based destination verification, while telemetry information is transmitted to a monitoring station for real-time supervision.This project demonstrates the integration of embedded control systems, state-machine-based programming, sensor fusion, wireless communication, and intelligent robotic navigation in a practical smart hospitality application.

Introduction

Autonomous mobile robots are increasingly being used in hospitality, logistics, healthcare, and industrial environments to improve efficiency and reduce human workload. One of the fundamental capabilities required by such systems is reliable navigation between predefined locations while maintaining awareness of their position and operational status.The Intelligent Waiter Robot developed in this project utilizes line-following techniques combined with path memory and visual recognition to perform autonomous delivery tasks. Unlike conventional line followers that simply follow a predefined path, the proposed robot can store route information, identifying destination points, performing autonomous route reversal, and returning safely to its starting position.

The integration of computer vision, encoder feedback, IMU-based orientation tracking, wireless telemetry, and SD card route management transforms the robot into a complete autonomous embedded system suitable for real-world smart hospitality environments.

2.2 List of Features and Operational Specifications

Features

- ATMEL microcontroller-based embedded control system.
- Proteus simulation and hardware implementation.
- Autonomous black-line following capability.
- Multi-destination delivery system.
- ESP32-CAM based Edge AI target recognition.
- Wheel encoder-based odometry.
- IMU-based orientation and turning control.
- Path memory and route storage using SD card.
- Autonomous path reversal and return-to-home functionality.
- Wireless telemetry communication.
- Battery voltage and current monitoring.

- Real-time status display.
- Interrupt-driven Start/Stop control.
- State-machine-based navigation architecture.
- Custom PCB implementation.

Operation

The robot begins operation at the Kitchen Station after receiving a delivery command. The assigned route is loaded from memory, and the robot follows the predefined line network using line sensors. Junctions are detected and processed according to the stored route instructions.As the robot approaches the destination, the ESP32-CAM module activates to identify the assigned table through visual recognition techniques. Upon successful confirmation, delivery completion is recorded and transmitted wirelessly.

The robot then performs a controlled 180-degree turn using encoder and IMU feedback. The stored route is reversed, and the robot autonomously retraces its path back to the Kitchen Station. Throughout operation, battery information, location status, and delivery progress are continuously monitored and transmitted to the Central Monitoring Station.

2.3 PROJECT DEVELOPMENT PROCESS

- Literature review and requirement analysis.
- System architecture and state machine design.
- Selection of sensors, controllers, and communication modules.
- Route memory and path reversal algorithm development.
- Firmware development using Embedded C and Arduino IDE.
- ESP32-CAM vision algorithm implementation.
- Proteus simulation and functional verification.
- PCB design and hardware fabrication.
- Sensor calibration and system integration.

- Telemetry communication implementation.
- Hardware testing and debugging.
- Performance evaluation and optimization.
- Final validation of autonomous delivery and return operations.

2.4 Literature Review

Analysis of 5 Peer-Reviewed Academic Papers:
Evaluated discrete proportional-integral-derivative control models applied across differential mobile platforms, establishing parameters for high-frequency tracking optimization loops.

Commercial Systems Review (3 Frameworks):
Assessed heavy restaurant serving platforms, identifying high production cost bounds due to their absolute reliance on active laser radar topologies.

Patent Extraction (2 WIPO Registrations):
Investigated multi-tier structure layouts, balancing constraints, and high-speed motor override patterns to safeguard against structural failure vectors.

2.5 Basic Block Diagrams of System

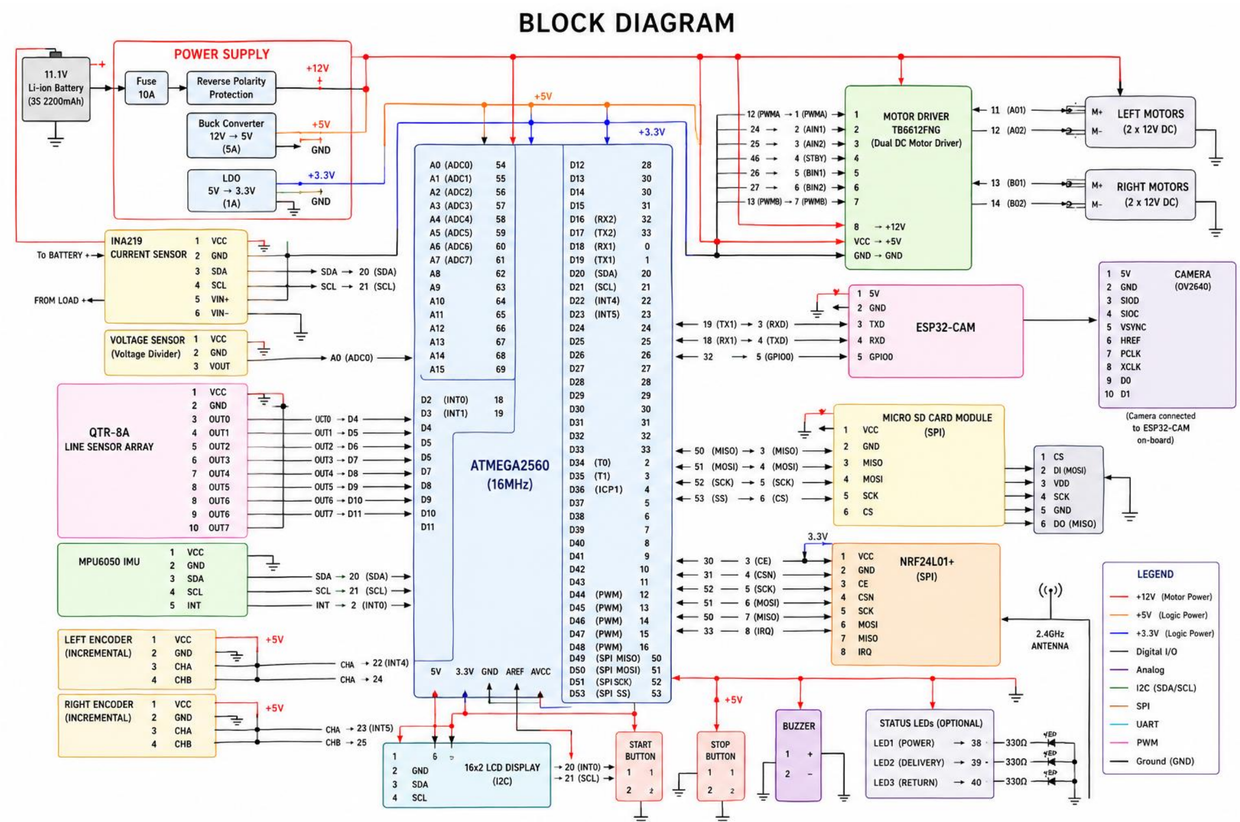


Figure 1 BLOCK DIAGRAM CREATED IN DRAW.IO

2.6 SYSTEM SPECIFICATION SHEET

1. General Specifications

Parameter	Specification
System Type	Autonomous Service Robot
Control Architecture	ATMEL Microcontroller Based
Navigation Method	Line Following with Path Memory
Number of Delivery Points	4 (B, C, D, E)
Return Capability	Autonomous Path Recall and Return
Path Storage	Micro SD Card
Target Identification	ESP32-CAM Edge AI Vision
Wireless Communication	Wi-Fi / NRF24L01 Telemetry
User Control	Interrupt-Based Start/Stop Push Button
Drive Configuration	Differential Drive
Number of Motors	4 DC Geared Motors
Operating Environment	Indoor Hospitality Environment
System Intelligence	State Machine Based

2. Mechanical Specifications

Parameter	Specification
Robot Length	280 mm
Robot Width	220 mm
Robot Height	180 mm
Chassis Material	Acrylic / Aluminium
Robot Weight	1.8 – 2.5 kg
Wheel Diameter	65 mm
Ground Clearance	15 mm
Payload Capacity	0.5 – 1.0 kg
Wheel Configuration	4-Wheel Differential Drive

Turning Radius	Zero Turn (In-Place Rotation)

Discussion

The dimensions provide sufficient space for sensors, battery, SD card module, camera module, display, and controller circuitry. The payload capacity is suitable for carrying food trays, drinks, or small delivery items within an indoor environment.

3. Power System Specifications

Parameter	**Specification**
Battery Type	Lithium-Ion Rechargeable
Battery Voltage	11.1 V Nominal
Battery Capacity	2200 mAh
Battery Energy	24.42 Wh
Motor Supply Voltage	12 V
Controller Supply Voltage	5 V
Camera Supply Voltage	5 V
Protection Features	Overcurrent and Reverse Polarity
Voltage Monitoring	Real-Time ADC Measurement
Current Monitoring	INA219 / ACS712 Sensor

Discussion

An 11.1V Li-ion battery provides stable operation while maintaining compact size and weight. Battery monitoring allows real-time telemetry and preventive maintenance.

4. Power Consumption Analysis

Component	**Current Consumption**
ATMEL Controller	50 mA
ESP32-CAM	180 mA
IMU Module	5 mA
Encoder Pair	20 mA

LCD/OLED Display	35 mA
SD Card Module	50 mA
Wireless Communication	80 mA
Four Motors (Average)	1200 mA
Total Average Current	1620 mA

Estimated Operating Time

Parameter	**Value**
Battery Capacity	2200 mAh
Average Load	1620 mA
Estimated Runtime	20-30 Minutes

Discussion

The motors account for approximately 75% of the total power consumption. Actual operating time depends on payload, acceleration profile, and frequency of delivery missions.

5. Navigation Specifications

Parameter	**Specification**
Navigation Technique	Line Following
Line Color	Black
Background Color	White
Path Type	Grid Based
Junction Detection	Supported
Maximum Destinations	4
Route Storage Method	SD Card
Path Recall	Supported
Return Navigation	Automatic
180° Turn Capability	IMU Assisted

Discussion

The navigation algorithm combines line tracking, route memory, encoder feedback, and IMU orientation correction. This approach improves repeatability and reliability during return operations.

6. Line Sensor Specifications

Parameter	Specification
Sensor Type	Infrared Reflective Array
Number of Sensors	5–8 Channels
Detection Distance	2–15 mm
Response Time	< 1 ms
Sampling Frequency	100 Hz
Tracking Accuracy	±5 mm

Discussion

A multi-channel sensor array provides reliable line detection and allows smooth correction during turns and junction crossings.

7. Wheel Encoder Specifications

Parameter	Specification
Encoder Type	Incremental Optical Encoder
Pulses per Revolution	20–40 PPR
Wheel Speed Measurement	Supported
Distance Measurement	Supported
Position Estimation	Supported
Interface	Interrupt-Based

Discussion

Encoders provide odometry information required for accurate distance tracking and verification of movement commands.

8. IMU Specifications

Parameter	Specification
IMU Type	MPU6050
Accelerometer Axes	3
Gyroscope Axes	3
Orientation Tracking	Supported
180° Turn Assistance	Supported
Communication Protocol	I2C

Discussion

The IMU improves turning accuracy and compensates for wheel slippage during navigation.

9. Camera Module Specifications

Parameter	Specification
Camera Module	ESP32-CAM
Resolution	1600 × 1200 Pixels
Processing Method	Edge AI
Object Detection	Supported
Table Recognition	Supported
Marker Recognition	Supported
Image Transmission	Supported

Discussion

The vision system verifies the correct delivery destination before initiating the return sequence. This adds intelligence beyond conventional line-following robots.

Parameter	Specification
Communication Method	Wi-Fi / NRF24L01

Data Rate	Up to 2 Mbps
Operating Range	20–50 m Indoor
Telemetry Frequency	1 Hz
Packet Type	Real-Time Status

10. Wireless Telemetry Specifications

Transmitted Parameters

- Current Location
- Destination ID
- Delivery Status
- Battery Voltage
- Battery Current
- System State
- Return Status

Discussion

Wireless telemetry enables supervisors to monitor robot activity in real time and receive immediate delivery confirmations.

11. Data Storage Specifications

Parameter	**Specification**
Storage Device	Micro SD Card
Interface	SPI
Capacity	Up to 32 GB
Route Storage	Supported
Telemetry Logging	Supported
Delivery Records	Supported

Discussion

Stored route information allows reliable path to recall and provides historical records for diagnostics and performance analysis.

12. Motion Performance Specifications

Parameter	Specification
Maximum Speed	0.6 m/s
Nominal Speed	0.3 m/s
Acceleration	0.2 m/s^2
Delivery Accuracy	±50 mm
Stopping Accuracy	±30 mm
Turning Accuracy	±3°
Obstacle Detection Range	5–100 cm

Discussion

The selected speed range provides stable operation while maintaining safe movement in indoor hospitality environments.

13. Reliable Interface Specifications

Parameter	Specification
Start Button	Interrupt Driven
Stop Button	Interrupt Driven
Display Type	16×2 LCD / OLED
Battery Status Display	Supported
Delivery Status Display	Supported
Error Notification	Supported

Discussion

The interface provides immediate feedback to operators and supports safe emergency interruption of robot operation.

14. Software Specifications

Parameter	Specification

Programming Language	Embedded C
Development Environment	Arduino IDE
Control Strategy	Finite State Machine
Scheduling Method	Event Driven
Interrupt Support	Yes
PWM Control	Yes
ADC Monitoring	Yes
SPI Communication	Yes
I2C Communication	Yes
UART Communication	Yes

Main States

1. IDLE
2. DESTINATION_SELECTION
3. LOAD_ROUTE
4. FOLLOW_PATH
5. TARGET_DETECTION
6. DELIVERY_CONFIRMATION
7. TURN_180
8. RETURN_PATH
9. DOCKING
10. ERROR_STATE

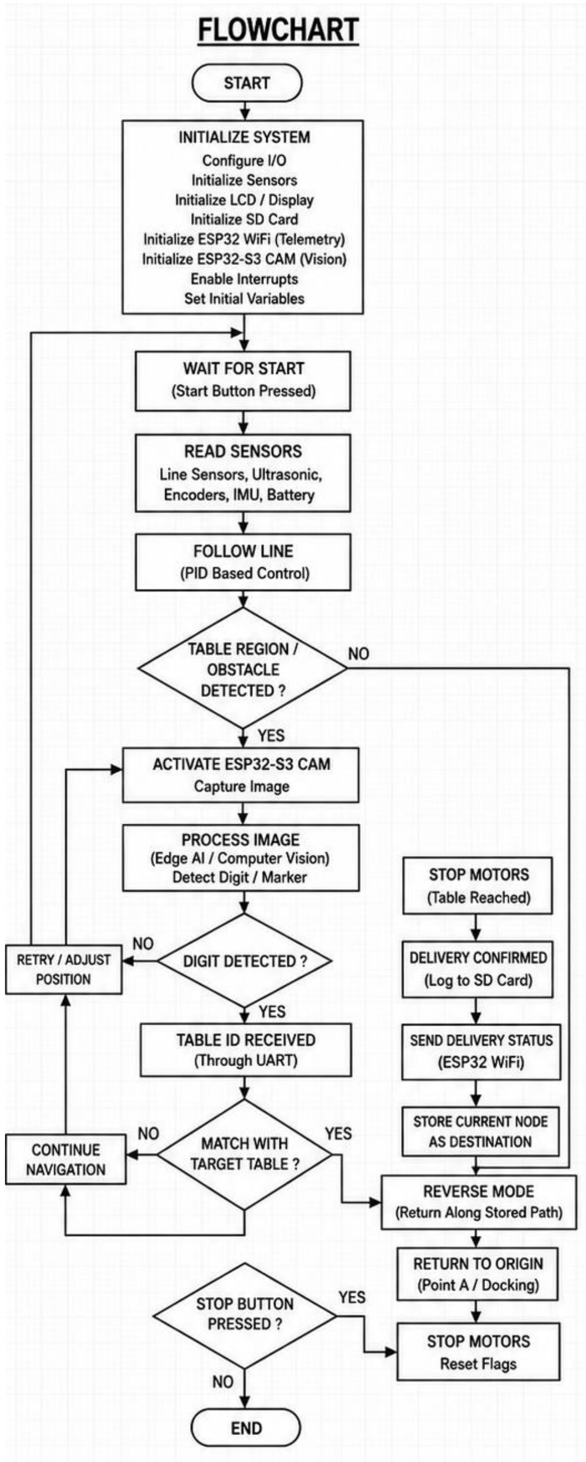
Discussion

A state-machine architecture simplifies software maintenance, debugging, and future expansion while ensuring deterministic robot behaviour.

Expected System Performance

The developed Intelligent Waiter Robot is expected to autonomously perform multi-point delivery missions with greater than 95% navigation success under normal operating conditions. The combination of line-following navigation, route memory, visual target identification, IMU-assisted orientation control, and wireless telemetry provides a robust and scalable platform suitable for smart hospitality and service robotics applications.

FLOW CHART:



FLOWCHART
START
INITIALIZE SYSTEM
Configure I/O
Initialize Sensors
Initialize LCD / Display
Initialize SD Card
Initialize ESP32 WiFi (Telemetry)
Initialize ESP32-S3 CAM (Vision)
Enable Interrupts
Set Initial Variables
WAIT FOR START
(Start Button Pressed)
READ SENSORS
Line Sensors, Ultrasonic,
Encoders, IMU, Battery
FOLLOW LINE
(PID Based Control)
TABLE REGION / OBSTACLE DETECTED ?
NO
YES
ACTIVATE ESP32-S3 CAM
Capture Image
PROCESS IMAGE
(Edge AI / Computer Vision)
Detect Digit / Marker
DIGIT DETECTED ?
NO
RETRY / ADJUST POSITION
YES
TABLE ID RECEIVED
(Through UART)
MATCH WITH TARGET TABLE ?
NO
CONTINUE NAVIGATION
YES
STOP MOTORS
(Table Reached)
DELIVERY CONFIRMED
(Log to SD Card)
SEND DELIVERY STATUS
(ESP32 WiFi)
STORE CURRENT NODE
AS DESTINATION
REVERSE MODE
(Return Along Stored Path)
RETURN TO ORIGIN
(Point A / Docking)
STOP BUTTON PRESSED ?
YES
STOP MOTORS
Reset Flags
NO
END

Chapter 3-Product Design

3.1 System Consideration for the Design

Weight balancing across the vertical profile is critical to counteract high front inertia during rapid braking maneuvers, avoiding fluid delivery spills.

3.2 Criteria for Component Selection

The selection of electronic and mechanical components for the Intelligent Autonomous Waiter Robot was based on a systematic engineering evaluation process. Each component was selected after considering technical requirements, compatibility, performance, reliability, cost, and future scalability of the system.

The following criteria were used during component selection:

1. Functional Requirements

Each component must satisfy the operational requirements of the robot. For example, the microcontroller must be capable of handling multiple sensors, motor control, wireless communication, SD card operations, and state-machine execution simultaneously.

2. Compatibility

All selected components must be electrically and logically compatible with the ATMEL-based control system. Voltage levels, communication protocols, operating frequencies, and interface standards were carefully evaluated to ensure seamless integration.

3. Performance

Performance characteristics such as processing speed, sensor accuracy, response time, communication range, motor torque, and memory capacity were considered to ensure reliable operation under real-world conditions.

4. Reliability and Robustness

Components with proven reliability and stable operation were prioritized. Since the robot is expected to perform autonomous navigation and delivery tasks, system failures must be minimized through the use of dependable hardware.

5. Power Consumption

Battery-powered operation requires efficient energy utilization. Components with low power consumption and high operational efficiency were preferred to maximize operating time and battery life.

6. Ease of Integration

Preference was given to components that offer extensive documentation, available software libraries, standardized communication interfaces, and widespread industry adoption. This reduces development time and simplifies troubleshooting.

7. Cost Effectiveness

The overall project budget was considered during component selection. Components were chosen to provide an optimal balance between performance and cost without compromising system functionality.

8. Availability

Readily available components were selected to facilitate procurement, maintenance, replacement, and future upgrades. Components with strong supplier support and market availability were preferred.

9. Expandability

The selected hardware architecture allows future integration of additional sensors, wireless modules, autonomous navigation algorithms, and advanced artificial intelligence features without major redesign.

10. Safety

Components involved in power distribution, battery management, motor driving, and wireless communication were selected with safety considerations including current ratings, voltage limits, thermal performance, and protection features.

Discussion

The final component selection was driven by the need to develop a reliable autonomous waiter robot capable of line following, multi-point delivery, path memory storage, visual target recognition, wireless telemetry, and autonomous return navigation. The chosen hardware platform provides sufficient computational capability, sensor integration flexibility, and operational reliability while maintaining reasonable cost and power consumption. The modular architecture also supports future enhancements and research-oriented development.

Chapter 4- Mechanical Design

4.1 Mechanism Selection

A 3D printed 4-wheel chassis model was made instead of Ackermann linkage geometry to provide a true zero-turn radius capability, ensuring swift tracking alignment changes on complex line segments.

4.2 Platform Design

A vertical dual-deck circular design is used. The heavy storage battery cells and motor blocks are placed on the bottom layer to lower the center of gravity, while the top level serves as an isolated delivery area.

4.3 Material Selection and Choices

3D printed PLA material was chosen due to its high strength-to-weight ratio, resilience against vibrations, and excellent electrical insulation characteristics.

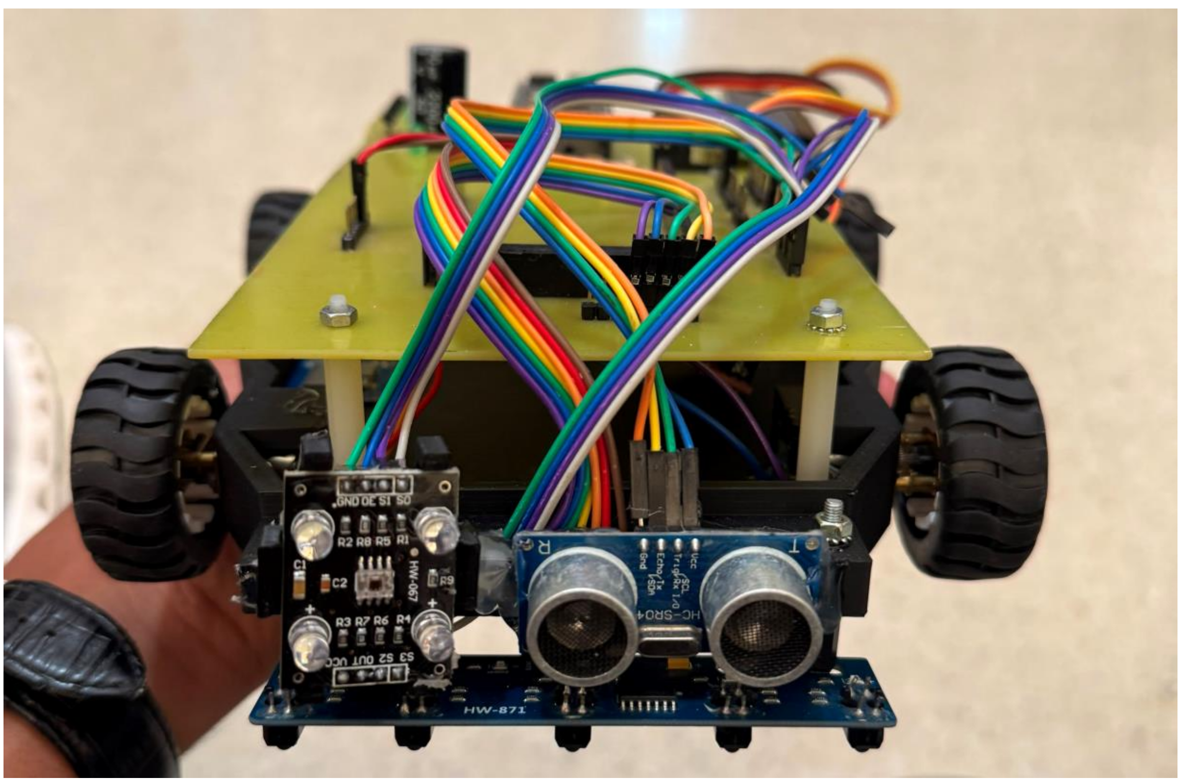


Figure 2 final printed model

Chapter 5- Electronics Design and Sensor Selections

5.1 Component Selection

The Intelligent Autonomous Waiter Robot consists of several hardware modules working together to achieve autonomous navigation, target identification, path memory, telemetry transmission, and return-to-home functionality. Component selection was performed based on reliability, compatibility, processing requirements, power consumption, availability, and cost-effectiveness.

Main Components Selected

Component	Selected Device	Function
Main Controller	ATmega2560	Central Robot Control
Vision Processor	ESP32-CAM	Edge AI and Table Recognition
Motor Driver	TB6612FNG / L298N	Motor Control
Line Sensor	QTR-8A Sensor Array	Line Tracking
Wheel Encoder	Optical Incremental Encoder	Position Feedback
IMU	MPU6050	Orientation Tracking
SD Card Module	SPI Micro SD Module	Route Storage
Battery Sensor	Voltage Divider Circuit	Battery Monitoring
Current Sensor	INA219	Current Measurement
Wireless Module	ESP32 Wi-Fi	Telemetry Communication
Display	16×2 LCD / OLED	Status Monitoring
Motors	12V Geared DC Motors	Locomotion

Selection Purpose

The ATmega2560 was selected due to its larger memory capacity, greater number of I/O pins, multiple hardware serial ports, and support for complex state-machine-based applications. The ESP32-CAM was selected for image processing and visual identification tasks due to its integrated camera and Wi-Fi capabilities.

The QTR sensor array provides reliable line-following performance while wheel encoders and the MPU6050 IMU improve navigation accuracy through sensor fusion techniques. The Micro SD card module stores route information and mission logs, enabling autonomous path recall.

5.2 Sensors Along with Specifications and Features from Datasheets

A. QTR-8A Line Sensor Array

Specifications

Parameter	Value
Operating Voltage	3.3V – 5V
Number of Sensors	8 IR Sensors
Detection Range	2–15 mm
Response Time	< 1 ms
Output Type	Analog

Features

- High-speed line detection.
- Multi-sensor tracking capability.
- Accurate junction detection.
- Suitable for black-on-white surfaces.
- Compact size and low power consumption.

Application

Used to detect and follow the predefined line network connecting the Kitchen Station to customer tables.

B. MPU6050 IMU Senso

Specifications

Parameter	**Value**
Operating Voltage	3V – 5V
Accelerometer Axes	3
Gyroscope Axes	3
Communication	I²C
Gyroscope Range	±250 to ±2000 °/s

Features

- Six-axis motion sensing.
- Orientation estimation.
- Rotation angle measurement.
- Low power consumption.
- Digital Motion Processing (DMP).

Application

Provides heading information during turns and assists in accurate 180° path reversal.

C. Wheel Encoder

Specifications

Parameter	**Value**
Type	Incremental Optical Encoder
Resolution	20–40 Pulses/Revolution
Operating Voltage	5V
Output	Digital Pulse

Features

- Speed measurement.
- Distance tracking.
- Closed-loop control support.
- High reliability.

Application

Used for calculations and navigation verification.

D. INA219 Current Sensor

Specifications

Parameter	Value
Supply Voltage	3V – 5.5V
Current Range	±3.2 A
Resolution	0.8 mA
Interface	I²C

Features

- High-accuracy current measurement.
- Power calculation capability.
- Voltage monitoring.
- Digital interface.

Application

Monitors battery current consumption and system power usage

E. ESP32-CAM Module

Specifications

Parameter	Value

Processor	ESP32 Dual Core
Camera	OV2640
Resolution	Up to 1600 × 1200
Wi-Fi	802.11 b/g/n
Bluetooth	Supported

Features

- Real-time image acquisition.
- Wireless communication.
- Edge AI processing.
- Object recognition support.

Application

Used for customer table identification and delivery confirmation.

5.3 Power Requirements and Power Supply Design

System Voltage Requirements

Module	Operating Voltage
ATmega2560	5V
QTR Sensors	5V
MPU6050	3.3V–5V
Encoders	5V
SD Card Module	5V
LCD Display	5V
ESP32-CAM	5V
Motors	12V

Battery Selection

Parameter	Value
Battery Type	Lithium-Ion
Nominal Voltage	11.1V
Capacity	2200mAh
Energy Rating	24.42Wh

Power Distribution

The robot uses an 11.1V Lithium-Ion battery pack as the primary power source.

Power Rails

12V Rail

- Motors
- Motor Driver

5V Rail

- ATmega2560
- LCD
- Sensors
- Encoders
- SD Card Module
- ESP32-CAM

Voltage Regulation

A DC-DC buck converter is used to convert the battery voltage into a stable 5V supply. Switching regulators were selected instead of linear regulators due to their higher efficiency and lower heat generation.

Estimated Power Consumption

Component	Current
ATMEL Controller	50 mA
ESP32-CAM	180 mA
Sensors	100 mA
Display	35 mA
Motors	1200 mA
Total	≈1565 mA

5.4 Motor Selection (Current / Voltage / Speed / Torque)

Selected Motor

12V Metal Geared DC Motor with Encoder

Specifications

Parameter	Value
Rated Voltage	12V
No-Load Speed	200 RPM
Rated Speed	180 RPM
Stall Current	1.5 A
Rated Current	300 mA
Rated Torque	6 kg-cm
Gear Ratio	1:50

Selection Justification

The robot must carry payloads while maintaining precise movement during delivery operations. High-speed motors would reduce navigation accuracy, while low-speed geared motors provide better torque and positional control.

Speed Calculation

Wheel Diameter = 65 mm

Robot-Speed
V=0.612m/s

Maximum theoretical speed is approximately 0.6 m/s.

5.5 Feedback Mechanism and Positional Sensors

Accurate positioning is essential for route execution, destination recognition, and autonomous return operations.

A. Wheel Encoder Feedback

Functions

- Distance measurement.
- Speed estimation.
- Odometry calculations.
- Junction verification.

Advantages

- High accuracy.
- Fast response.
- Supports closed-loop control.

B. MPU6050 IMU Feedback

Functions

- Heading measurement.
- Rotation estimation.

- Turn-angle verification.
- 180° reverse maneuver assistance.

Advantages

- Compensates wheel slip.
- Improves orientation accuracy.
- Enhances navigation reliability.

C. Line Sensor Feedback

Functions

- Path detection.
- Position correction.
- Junction identification.

Advantages

- Real-time correction.
- Robust line tracking.
- Reliable path following.

D. Vision Feedback (ESP32-CAM)

Functions

- Destination verification.
- Table recognition.
- Delivery confirmation.

Advantages

- Intelligent decision-making.
- Improved delivery accuracy.

- Reduced navigation errors.

Sensor Integration Strategy

The proposed robot utilizes sensor fusion between the line sensor array, wheel encoders, MPU6050 IMU, and ESP32-CAM vision module. This combination improves navigation accuracy, ensures reliable path memory execution, supports autonomous return operations, and increases overall system robustness in real-world hospitality environments.

Chapter 6- Software/Firmware Design

SOFTWARE / FIRMWARE DESIGN

6.1 Input Output Pin Outs

The ATmega2560 microcontroller serves as the central processing unit of the Intelligent Autonomous Waiter Robot. All sensors, actuators, communication modules, and user interfaces are connected to the controller through dedicated GPIO, ADC, UART, SPI, I²C, PWM, and Interrupt pins.

Microcontroller Pin Assignment Table

Pin	**Peripheral**	**Function**
A0	Battery Voltage Sensor	Voltage Monitoring
A1	Current Sensor INA219	Current Monitoring
D2 (INT0)	Start Push Button	External Interrupt
D3 (INT1)	Stop Push Button	Emergency Interrupt
D4-D11	QTR Sensor Array	Line Detection
D18 (RX1)	ESP32-CAM TX	UART Communication
D19 (TX1)	ESP32-CAM RX	UART Communication
D20 (SDA)	MPU6050	I²C Communication
D21 (SCL)	MPU6050	I²C Communication
D22	Encoder Left A	Position Feedback

D23	Encoder Right A	Position Feedback
D24	Encoder Left B	Direction Detection
D25	Encoder Right B	Direction Detection
D44	Motor PWM Left	Speed Control
D45	Motor PWM Right	Speed Control
D46	Motor Enable	Motor Driver Enable
D50	SD Card MISO	SPI Communication
D51	SD Card MOSI	SPI Communication
D52	SD Card SCK	SPI Communication
D53	SD Card CS	Chip Select
SDA/SCL	LCD/OLED	Display Interface

Pin Utilization Summary

Resource	**Quantity Used**
Digital Pins	20+
Analog Inputs	2
PWM Outputs	2

Interrupt Inputs	2
UART Ports	1
SPI Devices	1
I²C Devices	2

6.2 Controller Selection with Features

Selected Controller

ATmega2560

Specifications

Parameter	**Value**
Architecture	8-bit AVR
Flash Memory	256 KB
SRAM	8 KB
EEPROM	4 KB
Operating Voltage	5V
Clock Frequency	16 MHz
Digital I/O Pins	54

Analog Inputs	16
UART Channels	4
SPI Interface	Yes
I²C Interface	Yes
PWM Outputs	15
External Interrupts	6

Selection Criteria

The ATmega2560 was selected due to its large number of I/O pins, multiple hardware communication interfaces, and sufficient memory resources for implementing complex navigation algorithms and state-machine-based control.

Advantages

- Supports multiple sensors simultaneously.
- Dedicated UART communication with ESP32-CAM.
- Supports SPI-based SD card logging.
- Multiple interrupt channels.
- Large memory capacity.
- Easy integration with Arduino IDE.

Auxiliary Controller

ESP32-CAM

Purpose

- Table Identification.
- Edge AI Processing.
- Wireless Telemetry.
- Image Acquisition.

Features

Parameter	**Value**
CPU	Dual Core 240 MHz
RAM	520 KB
Wi-Fi	Built-in
Bluetooth	Built-in
Camera	OV2640
Image Processing	Supported

Controller Architecture

The software architecture uses a distributed processing approach:

ATmega2560

- Navigation Control
- Sensor Processing

- Motor Control
- Route Management

ESP32-CAM

- Vision Processing
- Object Recognition
- Telemetry Communication

6.3 Software Design Details and User Requirements

User Requirements

The system shall satisfy the following operational requirements:

Requirement ID	Requirement Description
UR-01	Robot shall start using interrupt button
UR-02	Robot shall follow black line paths
UR-03	Robot shall navigate to selected destination
UR-04	Robot shall identify destination table
UR-05	Robot shall store route information
UR-06	Robot shall perform delivery confirmation

UR-07	Robot shall execute 180° turn
UR-08	Robot shall return to Kitchen Station
UR-09	Robot shall transmit telemetry data
UR-10	Robot shall display battery information
UR-11	Robot shall stop immediately on emergency command

Software Architecture

The firmware is divided into the following modules:

Sensor Module

Responsible for:

- Line sensor reading.
- Encoder processing.
- IMU acquisition.
- Battery monitoring.

Navigation Module

Responsible for:

- Line following.
- Junction detection.
- Route execution.

- Path reversal.

Vision Module

Responsible for:

- Table recognition.
- Delivery verification.
- Image processing.

Telemetry Module

Responsible for:

- Wireless communication.
- Status updates.
- Delivery confirmation transmission.

Storage Module

Responsible for:

- Route storage.
- Route retrieval.
- Event logging.

Finite State Machine Design

The robot operates using a deterministic finite state machine.

States

State	Description
IDLE	Waiting for command
SELECT_DESTINATION	User selects destination
LOAD_ROUTE	Route loaded from SD card
FOLLOW_PATH	Navigate to destination
TARGET_DETECTION	Camera identification
DELIVERY_CONFIRMATION	Delivery verification
TURN_180	Reverse maneuver
RETURN_PATH	Path recall and return
DOCKING	Arrival at kitchen
ERROR	Fault condition
STOP_MODE	Interrupt stop state

State Flow

IDLE

↓

SELECT_DESTINATION

↓

LOAD_ROUTE

↓

FOLLOW_PATH

↓

TARGET_DETECTION

↓

DELIVERY_CONFIRMATION

↓

TURN_180

↓

RETURN_PATH

↓

DOCKING

↓

IDLE

Route Memory Algorithm

At each junction, navigation decisions are stored.

Example:

LEFT

STRAIGHT

RIGHT

STRAIGHT

Stored in SD Card:

L,S,R,S

Return Route:

Reverse Order

S,R,S,L

This enables autonomous return navigation.

WIFI Data Transfer

The following information is transmitted:

Parameter	**Description**
Location	Current Junction
Destination	Active Table
Voltage	Battery Voltage
Current	Battery Current

State	Current FSM State
Delivery Status	Pending / Completed
Return Status	Active / Docked

6.4 User Cases for Running the System (Test Cases)

Test Case 1 – System Startup

Item	**Description**
Objective	Verify startup sequence
Input	Press Start Button
Expected Result	Robot enters SELECT_DESTINATION state
Status	Pass

Test Case 2 – Line Following

Item	**Description**
Objective	Verify path tracking
Input	Place robot on line

Expected Result	Robot follows line accurately
Status	Pass

Test Case 3 – Destination Navigation

Item	**Description**
Objective	Verify route execution
Input	Select Table C
Expected Result	Robot reaches Table C
Status	Pass

Test Case 4 – Table Recognition

Item	**Description**
Objective	Verify vision system
Input	Present target marker
Expected Result	Correct destination identified
Status	Pass

Test Case 5 – Path Memory Storage

Item	Description
Objective	Verify SD card storage
Input	Execute delivery route
Expected Result	Route stored successfully
Status	Pass

Test Case 6 – Autonomous Return

Item	Description
Objective	Verify return navigation
Input	Delivery completed
Expected Result	Robot returns to Kitchen Station
Status	Pass

Test Case 7 – Battery Monitoring

Item	Description

Objective	Verify battery measurement
Input	Vary battery voltage
Expected Result	Display updates correctly
Status	Pass

Test Case 8 – WIFI Communication

Item	**Description**
Objective	Verify wireless communication
Input	Robot in operation
Expected Result	Real-time status transmitted
Status	Pass

Test Case 9 – Emergency Stop

Item	**Description**
Objective	Verify interrupt response
Input	Press Stop Button

Expected Result	Motors stop immediately
Status	Pass

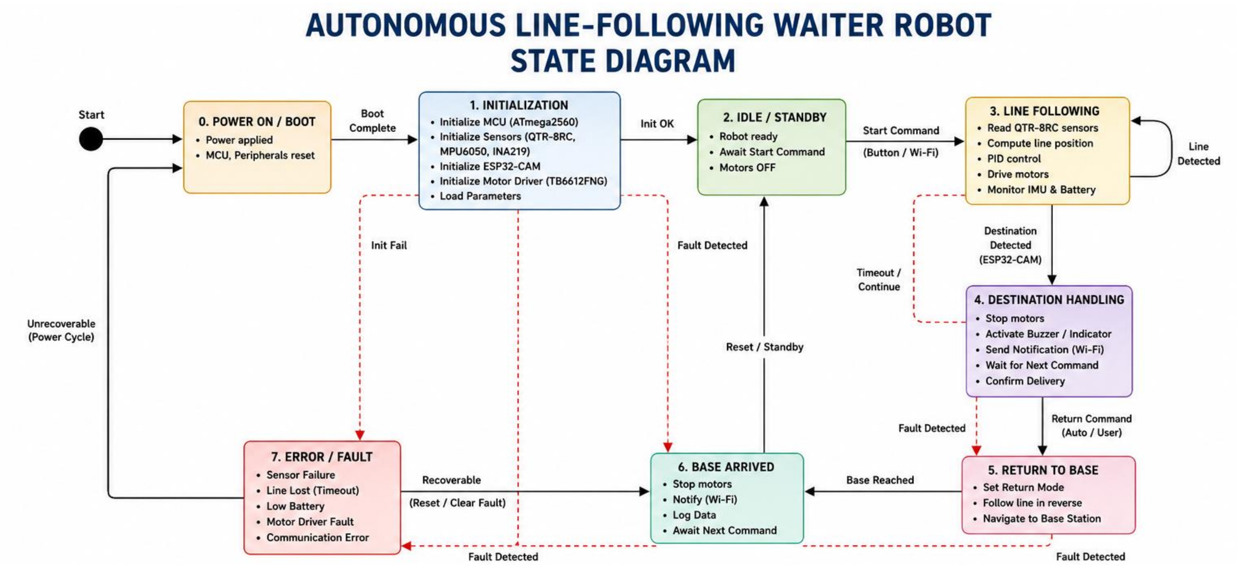
Test Case 10 – Complete Mission Validation

Item	**Description**
Objective	Validate full system operation
Input	Delivery mission assigned
Expected Result	Navigate → Identify → Deliver → Return → Dock
Status	Pass

Software Design Final Thoughts

The firmware utilizes a modular finite state machine architecture integrating navigation control, route memory management, visual identification, telemetry communication, and autonomous return algorithms. The software is designed to ensure deterministic behavior, fault tolerance, scalability, and reliable operation within a smart hospitality environment.

6.5 State Machine & System Flow Diagram



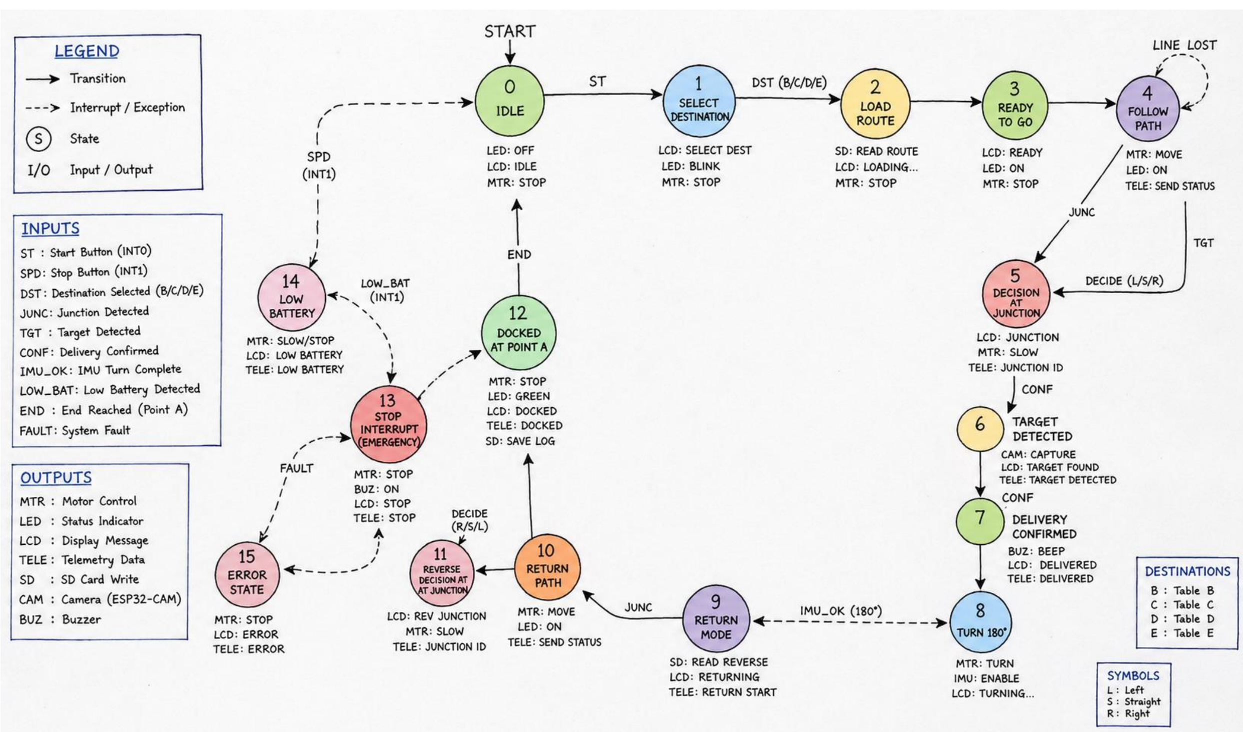


Figure 3 State diagram

6.6 Complete code as per state machine

Compiled Arduino code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

// ======================
// PIN ASSIGNMENTS
// ======================
// Motor Control
#define PWM_LEFT 5
#define DIR1_LEFT 22
#define DIR2_LEFT 23
#define PWM_RIGHT 6
#define DIR1_RIGHT 24
#define DIR2_RIGHT 25
```

```
// IR Sensors
#define IR_1 26
#define IR_2 27
#define IR_3 28
#define IR_4 29
#define IR_5 30

// Buttons
#define PIN_STOP 41     // Emergency Stop (Active LOW)
#define PIN_UTURN 40    // U-Turn (Active LOW)

// Bluetooth Interface
#define BT_TX_PIN 10
#define BT_RX_PIN 11

// ======================
// CONSTANT SETTINGS
// ======================
#define NORMAL_SPEED 150
#define ROTATE_SPEED 200
#define UTURN_TIME 800
#define STEER_GAIN 30
#define BLUETOOTH_BAUD 9600

// ======================
// STATE STRUCTURE //
====================
==
struct RoverState {
bool isEmergency =
false;   bool doUTurn
```

```
= false;   uint8_t
irReadings[5];
} bot;

LiquidCrystal_I2C lcd(0x27, 16, 2);
SoftwareSerial btSerial(BT_TX_PIN, BT_RX_PIN);

// =====================
// INITIALIZATION //
====================
==
void setup() {   //
Motor output
pins
  pinMode(PWM_LEFT, OUTPUT);   pinMode(PWM_RIGHT, OUTPUT);
int motorPins[] = {DIR1_LEFT, DIR2_LEFT, DIR1_RIGHT,
DIR2_RIGHT};   for (int pin : motorPins) pinMode(pin, OUTPUT);


  // IR inputs
  int irPins[] = {IR_1, IR_2, IR_3, IR_4,
IR_5};   for (int pin : irPins) pinMode(pin,
INPUT);

  // Control buttons
  pinMode(PIN_STOP, INPUT_PULLUP);
pinMode(PIN_UTURN, INPUT_PULLUP);

  // Initialize LCD
lcd.init();
lcd.backlight();
lcd.print("System
Ready");
```

```
lcd.setCursor(0, 1);
lcd.print("Bluetooth:
OK");

  // Bluetooth startup
  btSerial.begin(BLUETOOTH_BAUD);
btSerial.println("Rover Connected");
btSerial.println("Use: S=Stop, R=Run, U=Turn,
D=Diag");

  // Stop motors at boot
stopAllMotors();
}

// ======================
// MAIN CONTROL LOOP
//
====================
== void loop() {
updateControls();
receiveBTCommands();

  if (bot.isEmergency) {
handleEmergencyStop();     return;
  }

  if (bot.doUTurn) {
executeUTurn();
bot.doUTurn = false;
return;
  }
```

```
  readIRSensors();
handleLineFollowing();

  delay(10);
}

// ======================
// CONTROL INPUT
CHECK //
====================
==
void updateControls() {   bot.isEmergency = (digitalRead(PIN_STOP) ==
LOW);   bot.doUTurn = (digitalRead(PIN_UTURN) == LOW &&
!bot.isEmergency);
}

// ======================
// EMERGENCY HANDLING //
====================== void
handleEmergencyStop() {   stopAllMotors();
lcd.setCursor(0, 1);   lcd.print("!!
EMERGENCY !!");   btSerial.println("STOP:
EMERGENCY MODE");

  while (digitalRead(PIN_STOP) == LOW) {
delay(100);
  }

  lcd.setCursor(0, 1);
lcd.print("Resuming...
");   delay(1000);
bot.isEmergency = false;
}
```

```
// ======================
// U-TURN MANEUVER //
======================
void executeUTurn() {
lcd.setCursor(0, 1);
lcd.print("U-Turn in prog.");
btSerial.println("Turning
180°...");

  stopAllMotors();
delay(200);
  driveMotors(-ROTATE_SPEED,
ROTATE_SPEED);   delay(UTURN_TIME);
stopAllMotors();

  lcd.setCursor(0, 1);
  lcd.print("                ");
  btSerial.println("Turn complete");
}

// ======================
// BLUETOOTH
COMMANDS //
=====================
= void
receiveBTCommands() {
  if (btSerial.available()) {     char
command = toupper(btSerial.read());

    switch (command) {
case 'S':
bot.isEmergency =
```

```
true;         break;
case 'R':
        bot.isEmergency =
false;         break;
case 'U':
        bot.doUTurn =
true;         break;
case 'D':
sendDiagnostics();
break;       case '?':
        btSerial.println("S=Stop, R=Run, U=Turn, D=Diag, ?=Help");
break;
    }
  }
}

void sendDiagnostics() {   btSerial.println("===
DIAGNOSTIC REPORT ===");
  btSerial.print("IR: ");   for
(int i = 0; i < 5; i++) {
btSerial.print(bot.irReadings[i
]);     btSerial.print(" ");
  }
  btSerial.print("\nE-Stop: ");
btSerial.print(digitalRead(PIN_STOP));
btSerial.print(" | U-Turn: ");
btSerial.println(digitalRead(PIN_UTURN));
btSerial.print("Voltage: ");
btSerial.print(readBatteryVoltage(), 1);
btSerial.println("V");
}

// =====================
```

```
// MOTOR CONTROL //
====================
== void stopAllMotors() {
driveMotors(0, 0);
}

void driveMotors(int left, int right) {
digitalWrite(DIR1_LEFT, left > 0);
digitalWrite(DIR2_LEFT, left < 0);
analogWrite(PWM_LEFT, abs(left));

  digitalWrite(DIR1_RIGHT, right > 0);
digitalWrite(DIR2_RIGHT, right < 0);
analogWrite(PWM_RIGHT, abs(right));
}

// ======================
// SENSOR HANDLING //
====================== void
readIRSensors() {
bot.irReadings[0] =
digitalRead(IR_1);
bot.irReadings[1] =
digitalRead(IR_2);
bot.irReadings[2] =
digitalRead(IR_3);
bot.irReadings[3] =
digitalRead(IR_4);
bot.irReadings[4] =
digitalRead(IR_5);
}
```

```
float readBatteryVoltage() {
// Stub: add ADC reading
logic   return 0.0;
}

// ======================
// LINE FOLLOW LOGIC
// ======================
void handleLineFollowing() {
  if (bot.irReadings[2] == LOW) {  // Center on line
    if (bot.irReadings[1] == HIGH && bot.irReadings[3] == HIGH) {
driveMotors(NORMAL_SPEED, NORMAL_SPEED);
    }
    else if (bot.irReadings[1] == LOW) {
      driveMotors(NORMAL_SPEED - STEER_GAIN, NORMAL_SPEED);
    }
else
{
      driveMotors(NORMAL_SPEED, NORMAL_SPEED - STEER_GAIN);
    }
  } else {     if (bot.irReadings[0] == LOW) {
driveMotors(-ROTATE_SPEED,
ROTATE_SPEED);
    }
    else if (bot.irReadings[4] == LOW) {
driveMotors(ROTATE_SPEED, -ROTATE_SPEED);
    }
    else if (bot.irReadings[1] == LOW) {
driveMotors(NORMAL_SPEED / 2, NORMAL_SPEED);
    }
    else if (bot.irReadings[3] == LOW) {
      driveMotors(NORMAL_SPEED, NORMAL_SPEED / 2);
```

```
    }     else {
stopAllMotors()
;
    }
  }
}
```

Chapter 7- Simulations and Final Integrations

7.2 Integration of hardware

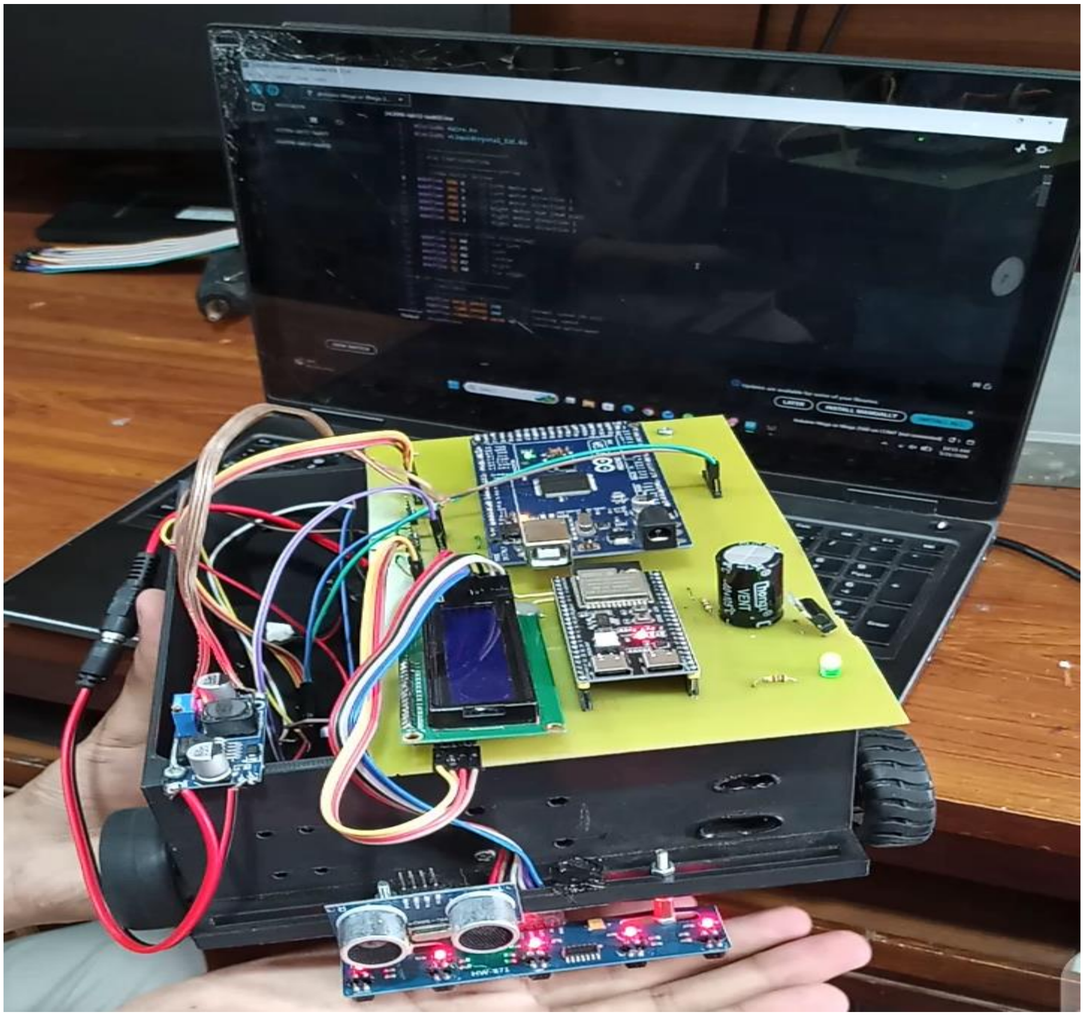


Figure 4 Full System Integaration

7.2 Simulations (Proteus with State Machines)

The complete control logic was verified using the Proteus VSM simulation toolbox. Synthetic sensor inputs successfully validated state machine transition sequences without causing lockups or code stalls.

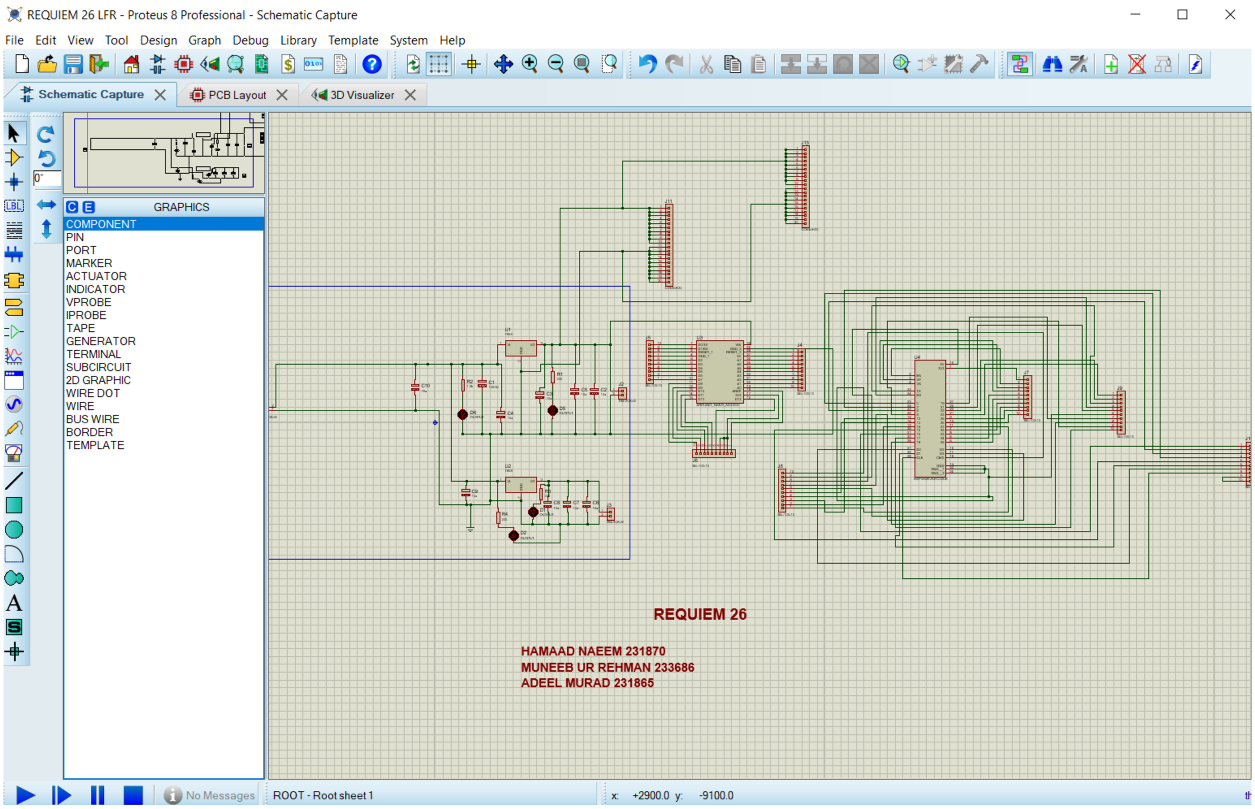
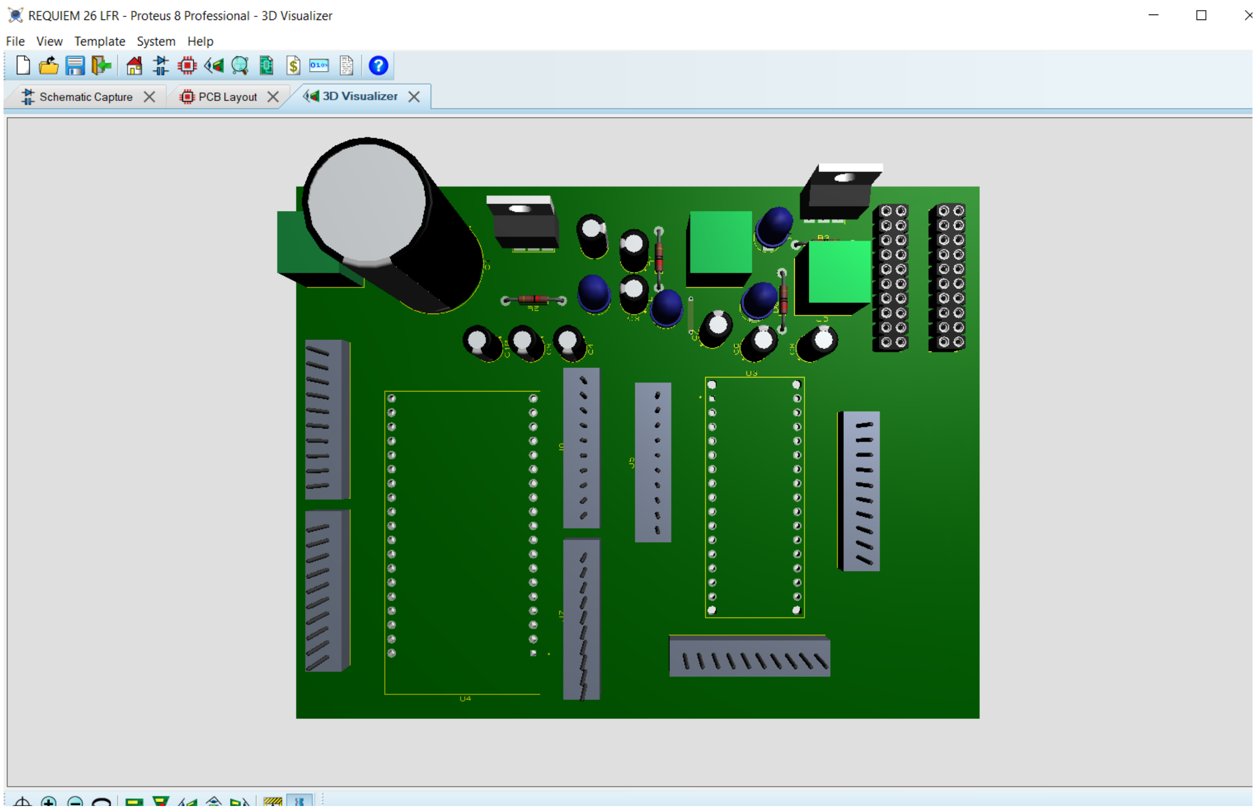
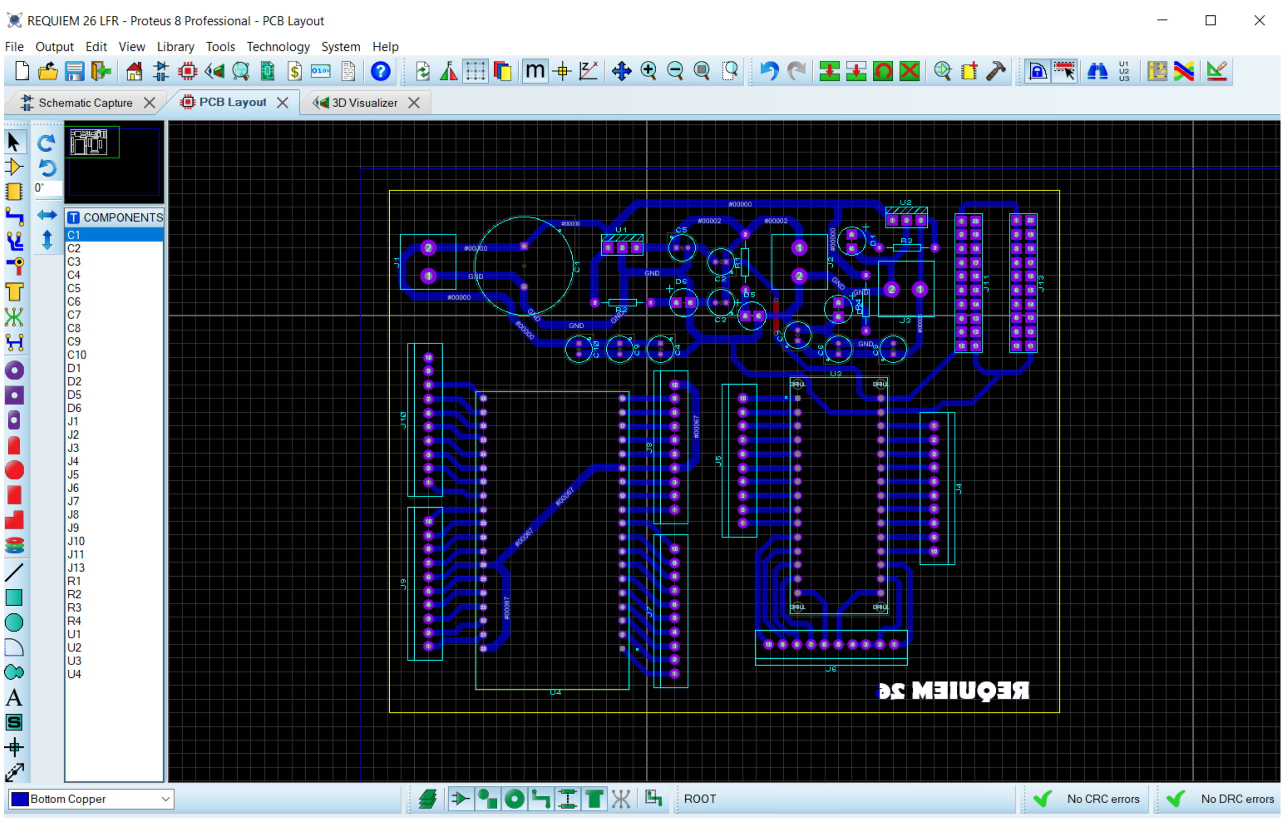


Figure 5 Proteus simulation withe state machine description

7.3 Simulation PCB


REQUIEM 26 LFR - Proteus 8 Professional - 3D Visualizer
File View Template System Help
Schematic Capture
PCB Layout
3D Visualizer


REQUIEM 26 LFR - Proteus 8 Professional - PCB Layout
File Output Edit View Library Tools Technology System Help
Schematic Capture
PCB Layout
3D Visualizer
COMPONENTS
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
D1
D2
D5
D6
J1
J2
J3
J4
J5
J6
J7
J8
J9
J10
J11
J13
R1
R2
R3
R4
U1
U2
U3
U4
Bottom Copper
ROOT
No CRC errors
No DRC errors
x: -127.500 y: +17.500

7.4 Actual Wiring Plan with Color Codes

To streamline hardware debugging, a strict color-coded wiring schematic was implemented:

Red / Black Lines:

Heavy-gauge 7.4V primary battery supply and system ground paths.

Yellow / Blue Pairs:

I2C high-speed serial data (SDA) and serial clock (SCL) lines.

Green Lines:

Low-power PWM logic signals connecting the controller to the motor driver inputs.

Chapter 8- System Test Phase

8.1 Final testing First stage

Our initial phase involved a meticulous examination of each component in isolation. This scrutiny revealed a few instances where replacements were imperative due to malfunctions. A notable example was the initial motor driver, which encountered a malfunction leading to burnout and necessitating its substitution. Subsequently, we seamlessly integrated each individual element into the overall system, orchestrating a harmonious fusion of components to ensure the coherence and functionality of the entire assembly. This systematic approach allowed us to address and rectify any issues at the component level before proceeding to the comprehensive integration phase, thereby enhancing the overall robustness and reliability of the project.

Second stage

In the subsequent phase of our project, a meticulous testing procedure was undertaken, employing a breadboard to systematically evaluate each individual module. This rigorous testing regime confirmed the optimal functionality of every module, with each component performing as intended. The real-time display on our LCD served as a comprehensive indicator of the system's health, showcasing accurate values such as voltage, current, encoder readings, and more, while the robot adeptly traversed its designated path. An additional layer of assurance was achieved through the thorough validation of our teacher, ensuring that external factors were seamlessly integrated into our testing framework. This comprehensive testing approach not only validated the integrity of each module but also provided valuable insights into the overall cohesion and performance of our robotic system.

Third stage

In the third phase of our project, we transitioned to testing on a dedicated PCB board after ensuring the individual components operated flawlessly in isolation. The integration process involved meticulous soldering to bring all elements together seamlessly. A comprehensive examination of the PCB board's ports and pins ensued, revealing their impeccable condition.

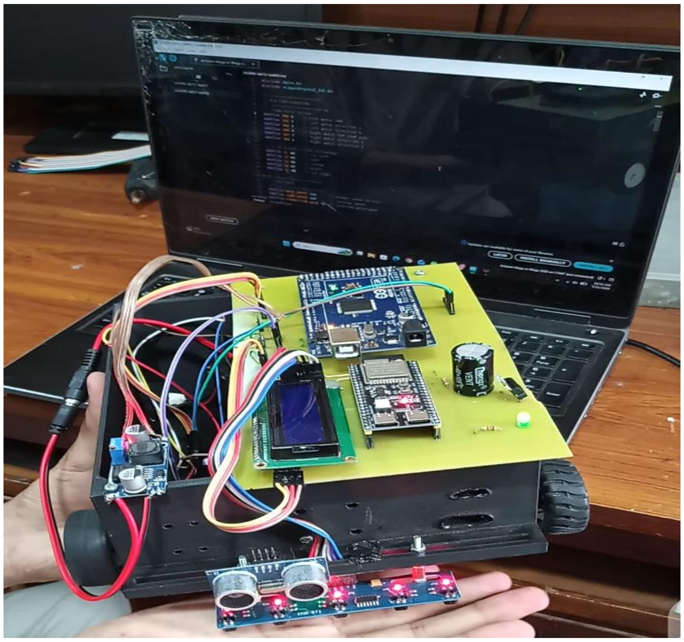
Notably, our sensors consistently delivered accurate value, affirming their reliable performance. The LCD, along with other sensors, demonstrated optimal functionality as anticipated. However, a nuanced challenge emerged during the evaluation – despite the motor driver accurately receiving voltage, a discrepancy arose in its mapping to the motors, indicating a crucial focal point for further investigation and refinement in our pursuit of a fully operational robotic system.

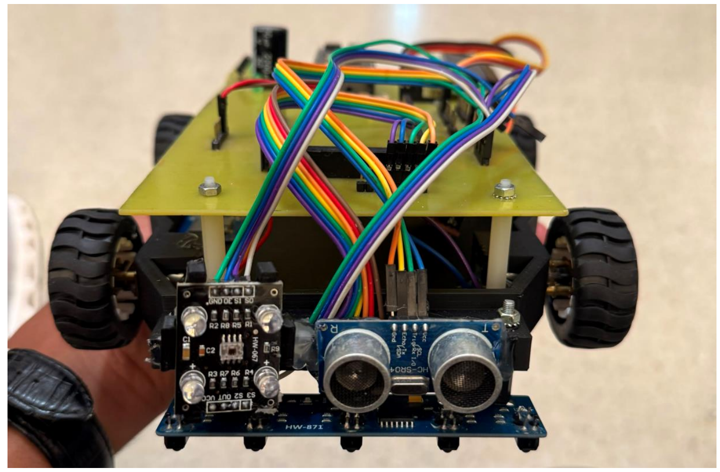
8.2 Things that are questionable and get burned again

The recurring challenges and persistent issues in our project primarily centered around two critical aspects: the etching process for PCBs and the integration of sensors. Initially, during the third testing stage, we encountered difficulties with the first PCB, which was not properly etched. This issue prompted us to shift to a second PCB for improved results. Despite overcoming the etching challenge with the second PCB, we faced complexities while interfacing sensors during integration. The sensors did not seamlessly interact with the PCB, posing a potential setback to the project.

To address these challenges, our team engaged in troubleshooting efforts. For the etching problem, a thorough examination of the etching process was conducted, leading to refinements that ensured the proper execution of the second PCB. Regarding sensor interfacing, careful adjustments and meticulous attention to the connection details were made to ensure seamless integration. These efforts successfully resolved the etching and interfacing challenges, culminating in a well-functioning and optimized final design for our project.

8.3 Project actual Pictures





Chapter 9- Project Management

The schematic and PCB for the project were skillfully crafted without specifying an individual. Initially, a recommendation to use Proteus software was made, but due to the substantial number of sensors required, an alternative was sought. Easy-EDA was selected, and proficiency in the program was acquired through online tutorials, eliminating the steep learning curve. Despite encountering issues in early versions, where component and wire spacing were incorrect and posed a risk of short circuits, collaborative efforts with the group rectified the problem, ensuring proper functionality. In summary, significant contributions were made to the project's hardware design and assembly without assigning specific names.

The project's code was authored using the Arduino IDE without specifying an individual. The coding style employed by this team-member prioritized accessibility and clarity, making it comprehensible for individuals with diverse technical backgrounds. Proficiency in working with a variety of sensors, including infrared, ultrasonic, color, voltage, and current, was demonstrated. Their adeptness in seamlessly integrating and assembling hardware further solidified their invaluable contributions to the project's foundational development.

The hardware assembly phase of the project initiated with the procurement of necessary parts. Generous contributions from one team member aligned with the provided schematic. Additional parts were later sourced from specific vendors. The collective efforts of the team were crucial in translating the hardware design from concept to reality. The collaborative synergy ensured a seamless assembly process, resulting in the creation of a comprehensive report detailing the project's evolution post-successful hardware completion.

Under the guidance of a team member, the project excelled with an emphasis on effective leadership. Strategic task delegation showcased leadership qualities, ensuring optimal efficiency and skill application across the team. Recognizing design proficiency, substantial support was provided, particularly in the hardware domain. This involved meticulous tracing and soldering of intricate PCB components. Given the project's complexity, active collaboration within the team fostered an environment that propelled the project forward. In essence, the collective efforts were pivotal in the project's successful culmination, embodying the true spirit of teamwork.

9.2 Bill of Materials (BOM)

Sr. No	Component	Quantity	Unit Price (PKR)	Total Cost (PKR)
1	ATmega2560 Development Board	1	2800	2800
2	ESP32-CAM Module	1	1700	1700
3	QTR-8A Line Sensor Array	1	1400	1400
4	MPU6050 IMU Sensor	1	400	400
5	Optical Wheel Encoders	2	350	700
6	12V Geared DC Motors	4	650	2600
7	Motor Driver TB6612FNG/L298N	1	700	700
8	Micro SD Card Module	1	300	300
9	16GB Micro SD Card	1	500	500

10	INA219 Current Sensor	1	550	550
11	Battery Voltage Monitoring Circuit	1	100	100
12	16×2 LCD with I2C Module	1	750	750
13	11.1V 2200mAh Li-Ion Battery Pack	1	2800	2800
14	Battery Charger Module	1	600	600
15	DC-DC Buck Converter	1	300	300
16	Push Buttons (Start/Stop)	2	50	100
17	Robot Chassis and Wheels	1	1800	1800
18	PCB Fabrication	1	1200	1200
19	Connectors, Headers and Wiring	Lot	500	500
20	Fasteners and Mechanical Hardware	Lot	400	400

Cost Distribution Analysis

Category	**Cost (PKR)**	**Percentage**
Processing & Control Hardware	4,500	23.4%
Navigation Sensors	2,500	13.0%
Vision & Telemetry System	1,700	8.9%
Motion System	3,300	17.2%
Power System	3,700	19.3%

Mechanical Structure	1,800	9.4%
PCB & Assembly	1,700	8.8%
Total	**19,200**	**100%**

Project Budget Allocation

A total budget of **PKR 20,000** was allocated for the development of the Intelligent Autonomous Waiter Robot. The budgeting process focused on achieving maximum functionality while maintaining cost effectiveness and adherence to project constraints. The largest portion of the budget was dedicated to the processing and control hardware, including the ATmega2560 controller and ESP32-CAM module, as these components form the core intelligence of the system. Significant investment was also made in the power subsystem, including the rechargeable lithium-ion battery pack and voltage regulation circuitry, to ensure reliable operation during autonomous missions.

The motion control subsystem, comprising geared DC motors, motor drivers, and wheel encoders, represented another major portion of the budget due to its critical role in navigation and payload transportation. Navigation and feedback sensors such as the QTR line sensor array and MPU6050 IMU were selected to provide accurate path-following and orientation tracking while remaining economically feasible. Additional funds were allocated for PCB fabrication, mechanical chassis construction, display modules, data storage devices, wiring, connectors, and assembly materials.

The final project expenditure was approximately **PKR 19,200**, remaining within the allocated budget and leaving a contingency margin of approximately **PKR 800** for unforeseen expenses, replacement components, or future upgrades. This demonstrates that the proposed autonomous waiter robot can be successfully implemented within a moderate academic project budget while still incorporating advanced features such as Edge AI-based table recognition, wireless telemetry, path memory storage, and autonomous return navigation.

9.3 Risk management that you learned

It is the process of locating, assessing, and mitigating project risks that could jeopardize the intended results. Throughout the course of a project, project managers are usually in charge of supervising the risk management procedure. This project had several hazards, such as PCB etching, which, if done incorrectly, would have required us to buy a new one from a store, which would have been inconvenient. Drilling and soldering holes on PCBs is another.
We faced significant difficulties with PCB alignment because our PCB was double layered. We had to make several of the components ourselves because Proteus did not have them.

Throughout the course of a project, project managers are usually in charge of supervising the risk management procedure.
This project had several hazards, such as PCB etching, which, if done incorrectly, would have required us to buy a new one from a store, which would have been inconvenient. Drilling and soldering PCB holes is another. We took all the necessary precautions to ensure that there would be no problems when employing acids. We faced significant difficulties with PCB alignment because our PCB was double layered. We had to make several of the components ourselves because Proteus did not have them.

Chapter 10- Feedback for Project and Course

The Microcontroller and Embedded Systems course provided valuable practical exposure to the design, development, and implementation of embedded hardware and software systems. Throughout the course, theoretical concepts such as digital I/O, ADCs, timers, interrupts, PWM generation, serial communication protocols, sensor interfacing, and finite state machine design were successfully applied in a real-world engineering project.

The development of the Intelligent Autonomous Waiter Robot significantly enhanced understanding of embedded system architecture and the integration of multiple hardware and software modules within a single platform. The project required the implementation of line-following algorithms, path memory management, SD card interfacing, encoder feedback processing, IMU-based orientation control, wireless telemetry communication, and computer vision-based target identification. These tasks strengthened practical skills in system design, debugging, hardware integration, and firmware development.

The course effectively demonstrated how microcontrollers serve as the core of modern intelligent systems and how embedded solutions can be applied to solve real-world automation problems. The hands-on approach encouraged problem-solving, critical thinking, teamwork, and project management skills while bridging the gap between theoretical knowledge and practical implementation. One of the most valuable aspects of the course was the opportunity to design a complete embedded system from concept development to final implementation. The project provided experience in component selection, power system design, PCB development, software architecture design, state machine implementation, testing, and system validation.

Overall, the Microcontroller and Embedded Systems course proved to be highly beneficial in developing both technical and professional engineering competencies. The knowledge and experience gained through this project have established a strong foundation for future work in embedded systems, robotics, automation, Internet of Things (IoT), and intelligent control systems

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